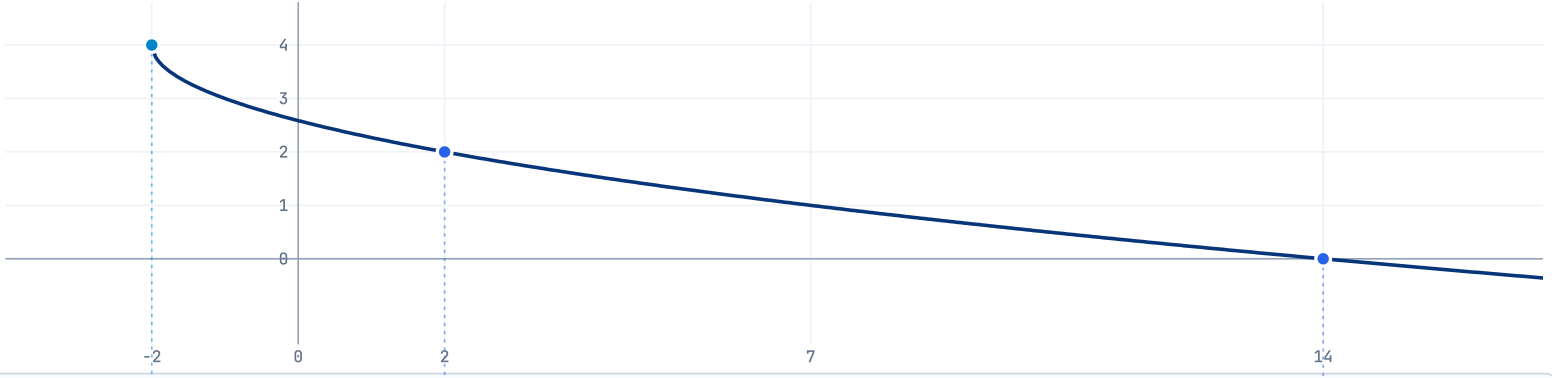
THE WORKED EXAMPLE

Profile of a transformed root

$f(x) = -\sqrt{(x+2)} + 4$

X ≥ -2 DOMAIN · √X PARENT



DOMAIN	undefined	defined, x ≥ −2
DIRECTION		decreasing (reflected)
CONCAVITY		concave up (reflected)

KEY POINTS

1 STARTS AT
$(-2, 4)$

2 POINT
$(2, 2)$

3 X-INTERCEPT
$(14, 0)$

GLOBAL

PARENT
$\sqrt{x}$
even index: start point, not centre

DOMAIN
$x \geq -2$
from $x + 2 \geq 0$

RANGE
$(-\infty, 4]$
the start is the highest point

TRANSFORMATIONS
left 2, reflect, up 4